

Estimating soybean yields from high-temporal-resolution multi-source data using deep learning

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ABSTRACT

Accurate and timely crop yield prediction is crucial for ensuring food security and maintaining stable agricultural markets, yet leveraging high-temporal-resolution, multi-source data for this purpose presents a big challenge in effectively capturing intricate variable interactions across time, especially during periods of extreme weather events. To address this issue, this study introduces an Attention and Graph Isomorphism Network-enhanced Bi-directional Long Short-Term Memory network (AGB-LSTM), which is specifically designed for estimating county-level soybean yield in the United States. The proposed model effectively integrated diverse high-temporal-resolution time-series data (5-days), including Near-Infrared Reflectance of Vegetation (NIRv), Solar-Included chlorophyll Fluorescence (SIF), and Gross Primary Productivity (GPP), along with environmental covariates. The AGB-LSTM model achieves an accuracy of $R^2 = 0.67$ and $rRMSE = 14.46\%$, which outperforms traditional and advanced machine learning methods tested, such as Random Forest (RF) ($R^2 = 0.52$, $rRMSE = 17.36\%$) and Transformer ($R^2 = 0.60$, $rRMSE = 15.80\%$). Furthermore, sensitivity analysis revealed the capability of the model for accurate and stable yield prediction 1 to 2 months before harvest. Notably, utilizing data with finer temporal resolution (5-day and monthly composite data) continuously improves performance. Specifically, the model performance is $R^2 = 0.67$ and $rRMSE = 14.46\%$ for the 5-day data, and $R^2 = 0.55$ and $rRMSE = 16.81\%$ for the 30-day data. We also highlight the robustness of the model under extreme climate events, maintaining strong performance with $R^2 = 0.50$ and $rRMSE = 21.32\%$. Finally, the validation against USDA reported yields for major North American soybean regions in 2023 show that AGB-LSTM well captures the spatial patterns of soybean yield. These findings underscore the AGB-LSTM model as a promising and effective method for yield estimation, showcasing large potential for global crop yield forecasting.

1. Introduction

Soybean is a vital global crop, widely used in food, animal feed and industrial products. The United States (U.S.) is a major producer and exporter, with soybeans covering about a quarter of the total crop area (Boryan et al., 2011). However, soybean yields exhibit notable inter-annual variability, influenced by factors such as climate change,

market demand, and extreme weather events like droughts and heavy rainfall. Accurate and timely monitoring of soybean growth and yield is crucial for ensuring a stable global supply and promoting sustainable agricultural practices (Liakos et al., 2018).

Traditional crop yield forecasting relies on either physical or statistical models. Physical models can dynamically simulate the process of crop growth and yield formation, thereby estimating yield (Peng et al.,

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2020a). However, they require high-quality data for calibration, which increases computational costs, especially for large-scale simulations. Additionally, these models often fail to account for extreme weather events, such as droughts and floods, which can lead to notable errors in yield estimation (Wang et al., 2023). Statistical models generally use environmental factors like temperature and precipitation as input variables. Recently, the integration of satellite remote sensing data, encompassing various spectral bands (optical, infrared, and microwave), has proven valuable in monitoring crop growth conditions and improving yield estimation (Guan et al., 2017; Zhang et al., 2025a, b; Liu et al., 2025). These data include surface reflectance, vegetation indices (VIs), leaf area index (LAI), fraction of photosynthetically active radiation (fPAR), and radar backscatter coefficients, among others (Peng et al., 2018).

Satellite-based Solar-Induced chlorophyll Fluorescence (SIF) has recently been demonstrated to be effective in capturing the spatial and temporal dynamics of terrestrial carbon uptake (Peng et al., 2020b). Guanter et al. (2014) were among the first to establish a link between SIF measurements and crop yield. In a county-level yield estimation study across the U.S., Peng et al. (2020b) integrated SIF with novel vegetation indices, such as Near-Infrared Reflectance of Vegetation (NIRv), and employed a suite of traditional machine learning algorithms, including LASSO, ridge regression, support vector regression (SVR), random forest (RF), and artificial neural networks (ANN) to estimate soybean yield. However, most existing studies rely on monthly or semi-monthly image synthesis. In contrast, relatively few studies focus on higher temporal resolutions, which can provide richer temporal information. With the increasing temporal resolution of remote sensing data, a key challenge arises in effectively extracting crop yield-related information from these high-frequency time series.

With the rapid advancements of artificial intelligence, machine learning has gained widespread use across diverse domains. Notable progress has been achieved in agricultural remote sensing research, including crop classification, yield estimation, and disease monitoring (Chen et al., 2016). For instance, deep learning techniques have demonstrated remarkable potential in simultaneous prediction of industrial parameters (Sukpancharoen et al., 2025), environmental contaminant monitoring (Rermborirak et al., 2025), and non-destructive quality assessment of agricultural products (Sarakum and Sukpancharoen, 2025). In the specific context of crop monitoring, machine learning is increasingly valued for its capacity to handle collinear variables and nonlinear relationships (Liakos et al., 2018). For instance, You et al. (2017) utilized MODIS data to develop a 2D histogram-based CNN model for predicting soybean yields across 11 U.S. states. Jiang et al. (2020) employed Long Short-Term Memory network (LSTM) to estimate county-level corn yields using meteorological and remote sensing data. Ma et al. (2021) proposed a Bayesian neural network (BNN) for predicting county-level corn yields and conducting uncertainty analysis by integrating optical remote sensing data, weather variables, and soil properties. Wang et al. (2023) improved the accuracy of wheat yield prediction by combining Convolutional Neural Networks (CNNs) and Gated Recurrent Units (GRUs) within a deep learning framework, utilizing remote sensing time series data. Tian et al. (2024) developed the Attention-based Multi-level Crop Network (AMCN) for estimating county-level wheat yield using both remote sensing and meteorological data. Mirhoseini Nejad et al. (2024) developed a soybean yield prediction method that fuses Convolutional Long Short-Term Memory (ConvLSTM), Three-Dimensional Convolutional Neural Network (3D-CNN), and Vision Transformer 1 (ViT). This model utilizes multispectral remote sensing data, combining the spatial hierarchical structure of 3D-CNN, the temporal modeling capability of ConvLSTM, and the ability of ViT to analyze global context, thereby capturing complex patterns in agricultural datasets. While traditional deep learning methods, particularly LSTM networks, effectively capture long-term dependencies in univariate time series, challenges arise when addressing multivariate time series with complex spatio-temporal relationships. As the

complexity of remote sensing data increases, especially with large-scale multivariate time series, a key challenge lies in effectively modeling these intricate non-linear interdependencies among variables (Ding and Yang, 2023).

In recent years, Graph Neural Networks (GNNs), an advanced class of deep learning models, have achieved notable success across various fields, including remote sensing data analysis. GNNs excel at processing graph-structured data by effectively modeling relationships between nodes and propagating information. This allows them to capture complex interactions and dependencies within a graph (Wu et al., 2020a). Graph Isomorphism Networks (GINs), a specialized type of GNN, are designed for accurate information aggregation and effective node representation learning, which enables them to effectively distinguish subtle differences in graph structures. In remote sensing applications that require fine-grained structural differentiation, GINs have demonstrated superior performance compared to traditional GNNs (Xu et al., 2019). Specifically, GINs excel at capturing the implicit correlation information between different variable features at different time nodes within time series data. Furthermore, GINs can dynamically assign weights to each time step, facilitating the detection and exploitation of feature interactions (Wu and Chen, 2024).

In this study, we proposed an Attention and Graph Isomorphism Network-enhanced Bi-directional Long Short-Term Memory network (AGB-LSTM) for estimating soybean yields at the county level in the U.S. AGB-LSTM integrates a GIN with an Attention Mechanism to enhance its representational capabilities. By leveraging GIN, the model effectively captures the intricate interactions between variables across different time steps within these high-temporal-resolution time series datasets. Furthermore, the incorporated attention mechanism dynamically identifies and weights the importance of different features at each time step, facilitating accurate modeling of complex temporal dynamics. Through comprehensive performance evaluation, we aim to demonstrate: (1) The superior performance of the AGB-LSTM model for county-level soybean yield prediction, highlighting its potential as a more accurate and timely alternative to traditional methods. (2) The value of integrating diverse data sources, such as remote sensing observations and environmental data, in improving yield prediction accuracy. (3) The robustness and effectiveness of the proposed model, especially its ability to maintain performance even when subjected to the stresses of extreme climate events.

This study aims to apply an AGB-LSTM model for crop yield estimation using high-temporal-resolution remote sensing sequence data and to evaluate its robustness under extreme climatic conditions. The structure of this paper is as follows: Section 2 introduces the study area, the datasets employed, and the preprocessing procedures. Section 3 presents the AGB-LSTM model and the experimental design. In Section 4, the model is trained and applied for yield prediction using high-temporal-resolution remote sensing sequences, the prediction accuracy of the AGB-LSTM model is assessed, and the effects of temporal range and sequence step length on prediction performance are investigated, along with the robustness of the model under extreme climate scenarios. Finally, Sections 5 and 6 provide discussion and conclusions, respectively.

2. Materials

2.1. Study area

This study focuses on soybean yield across 687 counties in 29 states of the central and eastern U. S. (Fig. 1). This region encompasses most of the soybean-growing areas in the U.S. for 2023, covering an elevation range from 0 to 2,600 m and a total area of approximately 4,562,160.25 square kilometers. Of this total area, approximately 332,940 square kilometers (7.3 %) is dedicated to soybean cultivation. The region spans from 75°W to 105°W longitude and 30°N to 45°N latitude, characterized by predominantly flat terrain, including fertile plains and favorable

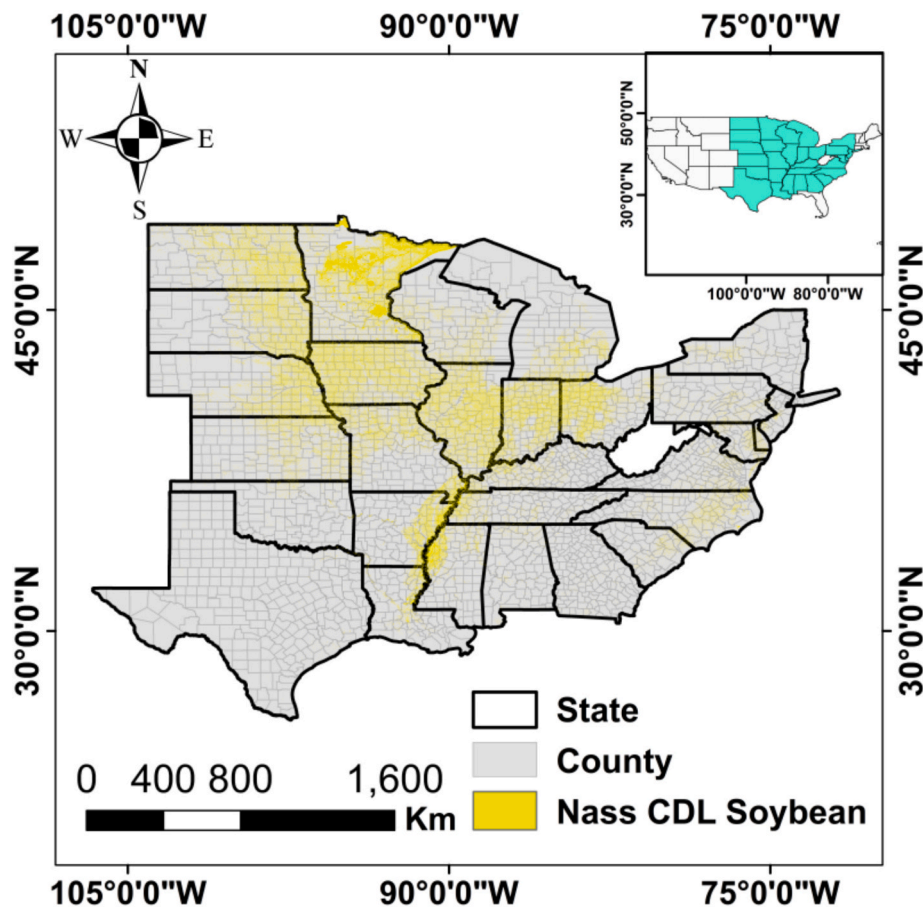


Fig. 1. Spatial map of soybean planting areas in the NASS CDL data in 2023. The black and gray lines represent the state and county boundaries, respectively. Comprising a total of 29 states: Alabama, Arkansas, Delaware, Georgia, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maryland, Michigan, Minnesota, Mississippi, Missouri, Nebraska, New Jersey, New York, North Carolina, North Dakota, Ohio, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Virginia, Wisconsin.

environments for soybean growth, making it an ideal agricultural region.

The study area experiences both temperate continental and subtropical humid climates, characterized by distinct seasonal variations. Annual precipitation ranges from 700 to 1,200 mm, with the majority occurring in spring and summer. This precipitation pattern closely aligns with the soybean planting and growing seasons. Soybeans are typically planted from early April to late May, with the main growth stages occurring from early May to late August. The pod-filling stage begins in late July, and harvest is mostly completed by September. This study specifically focuses on the critical growth and development stages from early May to the end of August.

2.2. Datasets and preprocessing

This study utilized multi-source data from six sources: Sentinel-2 satellite imagery, the MODIS LAI product (MCD15A3H, Collection v061), the GLDAS-2.1 Global Land Data Assimilation System (GLDAS) dataset, the CHIRPS global precipitation dataset, the GOSIF dataset (Li and Xiao, 2019), and the soybean yield data from the National Agricultural Statistics Service (NASS) of the U.S. Department of Agriculture (USDA). Details of the multi-source data used in this study are presented in Table 1. These data encompass the main growing season of soybeans, from early May to late August, for the years 2019 to 2023. Data from 2019 to 2022 were used for model training and validation, while data from 2023 served as an independent test dataset to evaluate the accuracy and generalization ability of the model. To some extent mitigate the impact of spatial heterogeneity and explore how variations in

Table 1
Characteristics of the multi-source data used in this study.

Dataset name	Temporal resolution	Spatial resolution	Interpretation	Feature name
Sentinel-2	5 days	10 m	Vegetation Index	NDVI, EVI, NIRv
GOSIF	8 days	0.05°	Solar-induced chlorophyll fluorescence (SIF)	SIF
GOSIF-GPP	8 days	0.05°	Gross Primary Productivity (GPP)	GPP
MCD15A3H	4 days	500 m	Leaf Area Index (LAI)	LAI
GLDAS-2.1	3 h	0.25°	Temperature and Soil moisture	Temp_min, Temp_mean, Temp_max, SM 0–10 cm, SM 10–40 cm
CHIRPS	1 day	0.05°	Precipitation	Prcp
USDA-CDL	1 year	30 m	Crop classification map	Crop map
USDA-Soybean Yield	1 year	County	County-level soybean yield data and coefficient of variation from USDA official statistics	Yield

hydrothermal patterns across different counties affect yield prediction and distribution, we ensured that every spatial unit (county) analyzed was represented in both the training and validation stages. This was

achieved by only including counties with at least one year of data in both datasets. This approach allows for a more reliable evaluation of the model's temporal generalization ability within the same spatial units and helps assess its capacity to capture yield variations caused by different hydrothermal conditions within specific spatial contexts. To facilitate county-level analysis, a series of preprocessing steps were applied to these datasets, which will be discussed in detail in the following sections.

2.2.1. Vegetation index data

Vis describe remote sensing vegetation characteristics, such as photosynthetic activity and canopy structure, and are widely used in cross-scale vegetation dynamics research. This study utilized the Harmonized Sentinel-2 MSI Level-2A satellite imagery to calculate vegetation indices that represent physiological characteristics of vegetation. The spatial resolution is 10 m, and the temporal resolution is 5 days. In this study, we employed a stringent cloud removal function (Yin et al., 2023). We created multi-layer masks to effectively remove pixels affected by clouds, snow, shadows, and cirrus clouds. Subsequently, a minimum value filter was applied within a 3×3 window to smooth the images and remove noisy pixels. The masked and smoothed images were then aggregated to a 30-meter spatial resolution. Next, we used the Cropland Data Layer (CDL) crop classification layer from USDA (<https://nassgeodata.gmu.edu/CropScape/>). The CDL data is an annual crop type classification product based on multiple satellite sources and employs a decision tree supervised classifier with a 30 m spatial resolution. For the Sentinel-2 images from 2019 to 2023, we applied a secondary mask based on the CDL layer, selecting pixels where the CDL value equals 5, which represents soybean fields. This ensured that the images we used encompassed all available soybean pixels while excluding irrelevant ones. Finally, we calculated the mean values of three indices—Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and NIRv—for all valid pixels within the county boundaries. The equations for these indices are presented in Eqs. (1)–(3):

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}} \quad (1)$$

$$\text{EVI} = \frac{2.5 \cdot (\text{NIR} - \text{Red})}{\text{NIR} + 6 \cdot \text{Red} - 7.5 \cdot \text{Blue} + 1} \quad (2)$$

$$\text{NIRv} = \text{NDVI} \cdot \text{NIR} \quad (3)$$

Using Eqs. (1)–(3), NIR represents the reflectance in the near-infrared band, Red represents the reflectance in the red band, and Blue represents the reflectance in the blue band.

For each county, this process is repeated with a time step of 5 days, resulting in time series curves of the mean values of the three vegetation indices for each county across a total of 25 time steps during the key growth stages of soybean.

2.2.2. SIF, GPP and LAI data

SIF is the energy flux emitted by plants within a few nanoseconds after light absorption during photosynthesis, and it directly provides a direct indicator of photosynthetic activity (Hao et al., 2022). At the ecosystem scale, SIF shows a strong linear relationship with GPP, which represents the total amount of carbon fixed by plants through photosynthesis. In this study, we employed a spatiotemporally continuous global GOSIF and GOSIF-derived GPP product, with a spatial resolution of 0.05° and a temporal resolution of 8 days. This product is derived from discrete SIF data observations obtained by OCO-2 and processed using machine learning methods. Compared to OCO-2, GOSIF offers higher spatiotemporal resolution. Additionally, GOSIF is continuous in both space and time, and it spans a longer time period (from 2000 to the present), which is more beneficial for monitoring crop growth at fine spatiotemporal scales over multiple years (Li and Xiao, 2019). The LAI

data used in this study comes from the MODIS MCD15A3H V061 LAI product dataset, which has a spatial resolution of 500 m and a temporal resolution of 4 days. We first derived time series curves of the mean values of SIF and GPP for each county with an 8-day time step, and then linearly interpolated these to a 5-day time step. For LAI, we calculated the mean LAI curve within a 5-day time window to ensure temporal alignment of the different parameters.

2.2.3. Environmental data

Temperature and soil moisture characteristics are derived from the GLDAS-2.1 dataset, which has a spatial resolution of 0.25° and a temporal resolution of 3 h. Temperature and soil moisture are commonly used to build crop yield prediction models, as they represent the most critical and direct thermal and hydrological conditions for crop growth and development in a given region. Precipitation data are sourced from the CHIRPS global precipitation dataset, which has a spatial resolution of 0.05° and a temporal resolution of 1 day. Precipitation and soil moisture are highly correlated to some extent, but short-term variations in precipitation intensity can significantly impact the final yield outcome. For temperature data, the maximum, minimum, and average temperatures are first calculated within a 5-day window. The temperatures are then converted from Kelvin to Celsius, resulting in time series data for the maximum, minimum, and average temperatures for each county with a 5-day time step. For soil moisture data, the soil moisture of the surface layer (0–10 cm) and the 10–40 cm depth layer is first calculated within a 5-day window, resulting in soil moisture time series data at different soil depths for each county with a 5-day time interval. For precipitation data, the cumulative daily average precipitation within a 5-day window is computed, yielding the time series of precipitation data for each county with a 5-day time step.

2.2.4. Soybean yield benchmark data

We obtained the county-level soybean yield data and their coefficients of variation for a total of 687 counties across 29 U.S. states from the USDA NASS QuickStats database (<https://quickstats.nass.usda.gov/>) for the years 2019 to 2023. As in Eq. (4), we estimated the reasonable range of yield predictions within a 95 % confidence interval (CI), which will be presented in the scatter plot:

$$\text{CI} = \frac{|\mu \pm (1.96\mu \cdot \frac{\text{CV}}{100}) - \mu|}{\mu} \times 100\% \quad (4)$$

In this context, μ represents the actual yield data as predicted by the USDA, while CV refers to its coefficient of variation.

As in Fig. 2, we applied the Savitzky-Golay filter to all time-series data for noise reduction (Chen et al., 2004). This method effectively mitigates the impact of noise on the signal while preserving the local trends and features of the signal. Finally, during the training process, we normalized the time-series data for all features independently, ensuring that the scales of different features are unified. This improves the stability and convergence speed of subsequent model training and prevents imbalanced effects on the model caused by features with different scales.

This study divides the time series data from May 1 to August 29 for each year from 2019 to 2022 into training and validation datasets in an 80 %: 20 % ratio. This split is used for model training and hyperparameter tuning, including dropout rate, learning rate, batch size, the size of fully connected layers, the number of hidden layers, and the number of hidden units, among others. The data from 2023 serves as an independent test dataset to evaluate the generalization performance of the model. Before yield estimation, correlation and significance analyses are conducted between all feature data at each time step and yield. This helps identify the most relevant features and time periods for yield prediction. The model is then trained using the selected features and time periods. The accuracy of various feature combinations is evaluated on the validation dataset to determine the optimal combination. For the deep learning model, 100 random parameter combinations are tested for

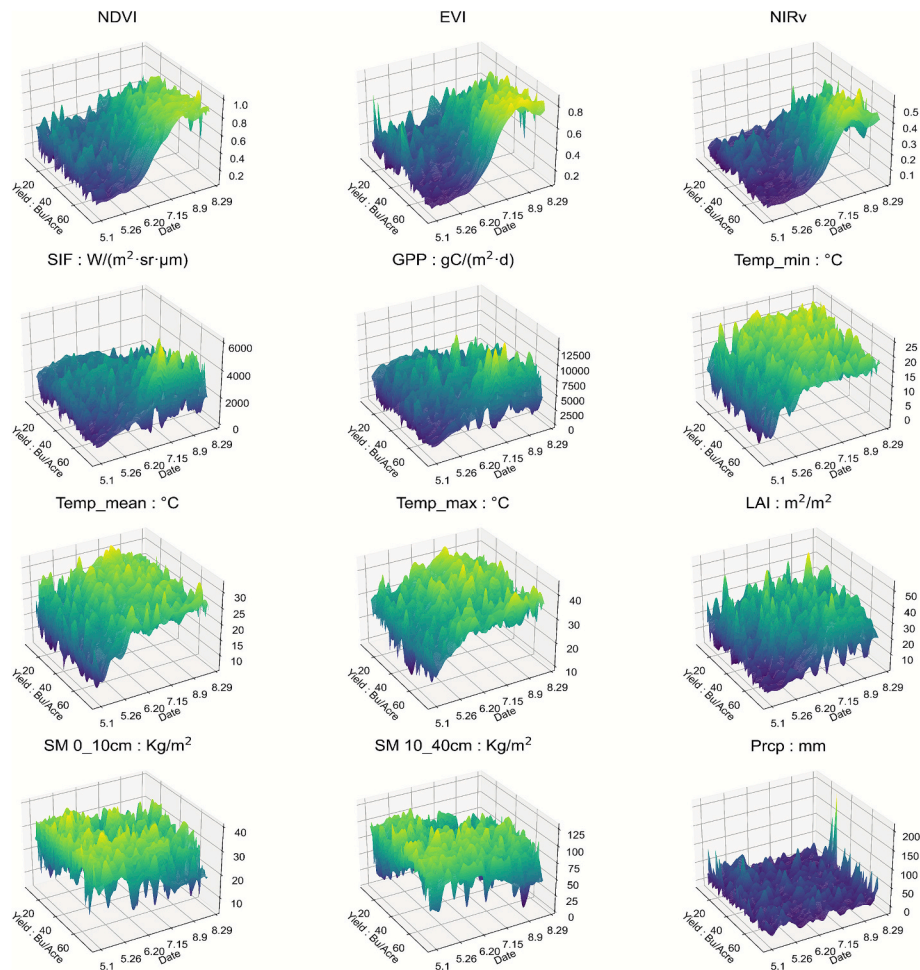


Fig. 2. Time series of different features. The x-axis represents yield, the y-axis represents date, and the z-axis represents the value of the features. The depth of the color represents the magnitude of the z-axis value.

each feature combination. The specific parameter names and tuning results are detailed in Table D in the [Supplementary Materials](#). The best parameters are determined based on the accuracy on the validation set, and their performance is then evaluated on the independent test dataset. For the machine learning model, five-fold cross-validation is employed to evaluate all parameter combinations. The optimal parameter set is selected based on the mean performance across the five validation folds. The final accuracy of the model is then assessed on the independent test dataset. This study used a paired *t*-test with a one-tailed hypothesis to determine whether the prediction error of the proposed model is significantly smaller than that of other models.

Based on the best model, we conducted in-season yield predictions over different time periods. Specifically, while keeping the starting dates of the sequence data constant (May 1 to August 29), we progressively advanced the end date of the period. For each time frame, we performed independent training and prediction. This approach aimed to investigate the capability of the model for early estimation of soybean yield. Furthermore, we averaged the sequence data over different time periods (e.g., one decade, half a month, and one month) to generate sequences with varying time steps. Subsequently, we evaluated the impact of these different time steps on the prediction accuracy for the test dataset.

In addition, we developed a Python version of the RCLimDex extreme climate event detection model, based on climate data ([Zhang and Yang, 2004](#)). Utilizing all sample data from 2019 to 2022 as a baseline, we calculated 26 extreme climate indices for 2023, including maximum daily temperature (TXx), minimum daily temperature (TNn), total precipitation (TR), number of days with precipitation ≥ 10 mm (R10mm),

consecutive dry days (CDD), and the 99th percentile of daily precipitation (R99p), among others. Extreme climate events were identified using the percentile threshold method ([Zhang and Yang, 2004](#)). The abbreviations, definitions, and threshold conditions for these indices are provided in [Supplementary Material Table A](#). To evaluate the ability of the model to predict crop yield under unseen extreme conditions, we classified the samples in the 2023 test set into those experiencing extreme climate events and those not experiencing extreme climate events. We then analyzed the prediction results separately for these two groups to assess the impact of extreme climate events on yield prediction accuracy and the generalization and stability of our model under these conditions.

2.3. Deep learning model development

Our research addresses this by employing a novel model, AGB-LSTM, which integrates three components: a Graph Isomorphism Network (GIN), an attention mechanism, and a Bidirectional Long Short-Term Memory network (Bi-LSTM).

The GIN is a deep learning architecture designed for graph-structured data. It excels at capturing both node features and topological information. Its key strength lies in a unique aggregation mechanism that enhances message-passing and information aggregation between nodes, resulting in strong expressive and discriminative power. Unlike traditional graph neural networks, GIN improves feature extraction by effectively aggregating information from each node and its neighbors ([Xu et al., 2019](#)).

GIN comprises two main components: the node embedding generator and the graph embedding generator. The node update and graph embedding processes of GIN are presented in Eqs. (5), (6), (7):

$$h_v^{(l)} = \text{MLP}^{(l)}((1 + \epsilon^{(l)})h_v^{(l-1)} + \sum_{u \in N(v)} h_u^{(l-1)}) \quad (5)$$

$$h_G = \text{CONCAT}(h^{(l)} | l = 0, 1, \dots, L) \quad (6)$$

$$h^{(l)} = \text{READOUT}(h_v^{(l)} | v \in G) \quad (7)$$

The embedding vector of the node v at the l -th layer is denoted as $h_v^{(l)}$, and at $l = 0$, it represents the input feature vector of node v . $N(v)$ refers to the set of neighbors of node v , while $\text{MLP}^{(l)}$ is a multilayer perceptron, and $\epsilon^{(l)}$ are the training parameters at the l -th layer. Where h_G represents the graph embedding, CONCAT is the concatenation operation, and READOUT denotes the pooling of node embeddings from each layer.

The attention mechanism can dynamically learn the weights of different time points (Vaswani et al., 2017), which helps to emphasize the most crucial time periods for soybean yield prediction. Bi-LSTM, on the other hand, captures the dependencies between past and future information (Schuster and Paliwal, 1997), enabling the model to learn the impact of factors such as crop growth and climate change over a broader time window. The model aims to capture dynamic features related to crop growth and yield from time series data extracted from remote

sensing images and predict soybean yield by leveraging graph structure learning and sequence modeling. As in Fig. 3, the model comprises three core modules: the GIN module, the Attention Mechanism module, and the Bi-LSTM module.

Specifically, the input layer accepts time series data extracted from remote sensing images. The input data comprises multi-dimensional features, including VIs, SIF, GPP, and environmental factors. Data for each time step is represented as a multi-dimensional vector. Specifically, the model expects to receive as input a tensor of shape (n, c) , where n is the number of nodes in the input graph (also referred to as the sequence length), and c is the dimension of the initial node feature vector. The input time series data is then transformed into a graph structure, and the relationships between time steps are learned using a GIN. More precisely, we use an adjacency matrix to build the model input sequence data into a graph structure. Here, each time step in the sequence data becomes a node, and edges connect different time step nodes based on their corresponding multi-dimensional features. For this graph-structured data, the Graph Isomorphism Network utilizes a Multi-Layer Perceptron (MLP) with ReLU as the activation function to perform non-linear transformations during the aggregation and update steps of the graph convolution. The output consists of the updated node embeddings, which capture the node's local graph structure and feature information. An attention mechanism is employed to calculate the feature weights for each time step, dynamically adjusting the contribution of the input data, thereby enhancing the focus on key moments.

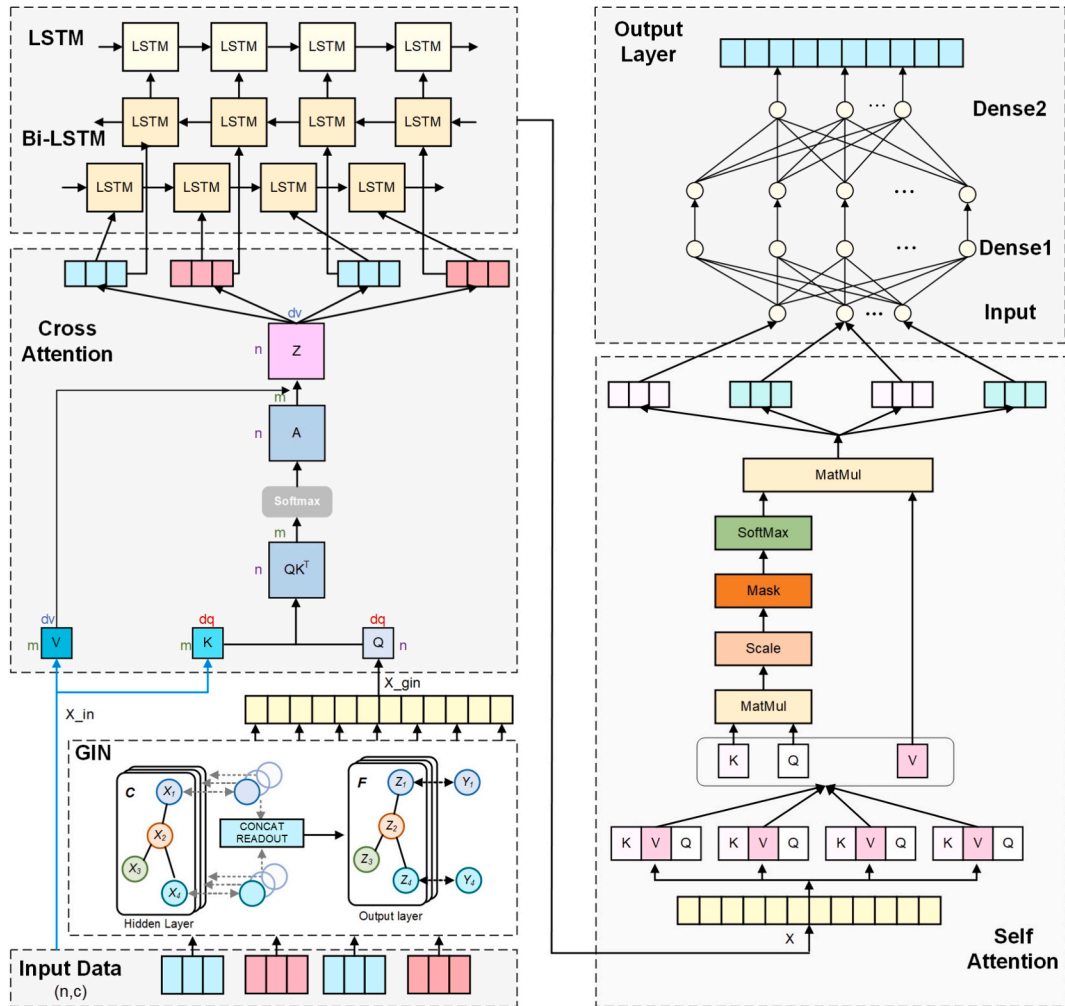


Fig. 3. Network structure of the AGB-LSTM model for soybean yield estimation. In the diagram of the GIN, C represents the hidden layer, where input features (X_1-X_4) are processed. F is the output layer, which produces final predictions. X_1-X_4 are the input features of nodes. Z_1-Z_4 are the transformed features in the hidden layer after aggregation. Y_1-Y_4 are the output predictions of the network for each node.

Here, we utilize both a cross-attention layer and a self-attention layer. The cross-attention layer takes x_{in} (original features) and x_{gin} (features processed by GIN) as inputs, computes attention scores, and generates a weighted representation. This enables the model to selectively attend to the graph structure information most relevant to the original input features, creating a combined sequential representation. Subsequently, this combined representation is processed by a BiLSTM layer followed by an LSTM layer. The self-attention layer then receives the output sequence from the LSTM layer (denoted as x). A Bi-LSTM network is employed to model the time series data, capturing long-term dependencies from both past and future time steps. This layer utilizes a Bi-LSTM network. The output is the sequence of hidden states at each position, rather than the final state. On top of the Bi-LSTM, we further stacked a standard unidirectional LSTM layer to further process the sequence generated by the Bi-LSTM, designed to model more complex sequence patterns. Adding self-attention at this stage serves to further refine the sequence representation. It allows the model to directly establish connections between distant positions within the sequence and assign different importance weights to different time steps, thereby producing a sequence representation with enhanced awareness of global context information. The output features are further processed by a dense layer to extract comprehensive features related to soybean yield. Specifically, the Flatten layer converts the output tensor of the self-attention layer into a flattened vector for each input sample. A fully connected (Dense) layer then performs a linear transformation, serving as a hidden layer used to learn high-order feature combinations from the flattened representation. Dropout regularization is employed to prevent overfitting during training by randomly setting a fraction of neuron activations to zero. The final output layer is a Dense layer with a single neuron, used for performing the regression task.

For the model comparison and validation, we employed three conventional machine learning algorithms for performance evaluation: Linear Regression (LR), Random Forest (RF), XGBoost. Before inputting the data into these traditional machine learning models, we performed a series of key data preprocessing steps. Since traditional linear and tree-based models typically handle fixed-length feature vectors rather than variable-length sequence data, we needed to reformat the enhanced time series data. This involved merging the temporal features with the original features at each time step. For each independent input sample (a time series containing n time steps), we applied a Flatten operation to the features across all time steps. If the data has 'a' features per time step, a sequence with 'n' time steps is flattened into a long vector of length $n \times a$. Following this feature merging and flattening process, each original sequence sample was transformed into a single, high-dimensional feature vector. This allows machine learning models like LR, RF, and XGBoost to directly use this flattened feature data for training and prediction. Furthermore, we conducted two ablation studies by creating two ablated variants of our proposed model: one without the Graph Isomorphism Network (GIN) module (AB-LSTM), and another without the attention mechanism module (Bi-LSTM). We also included a Transformer-based model as a comparative baseline for yield prediction, resulting in a total of three control groups used to evaluate the effectiveness of our proposed model. The Transformer model is a neural network architecture that uses self-attention mechanisms instead of traditional sequential processing (like in RNNs or LSTMs) to weigh the importance of different parts of an input sequence in parallel. There are many advanced studies that have demonstrated the effectiveness of transformers in crop yield prediction. For example, Guo et al. (2024) proposed a Transformer-based deep learning framework to estimate winter wheat yield in the Guanzhong Plain using remote sensing multivariate variables, namely LAI, fPAR, and vegetation temperature condition index (VTCI).

2.4. Evaluation of model performance

As in Eqs. (8)–(13), this study evaluated the prediction accuracy of

the model using the coefficient of determination (R^2), relative root mean square error (rRMSE), mean absolute percentage error (MAPE), relative bias (rBias), and standard error (SE).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y'_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

$$rRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y'_i)^2}}{\bar{y}} \quad (9)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - y'_i|}{y_i} \quad (10)$$

$$rBias = \frac{\sum_{i=1}^n y_i - \sum_{i=1}^n y'_i}{\sum_{i=1}^n y_i} \quad (11)$$

$$SE = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n(n-1)}} \quad (12)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y'_i| \quad (13)$$

here, y_i denotes the official soybean yield value published by the USDA, y'_i refers to the predicted value from the model, and $\bar{y}$ represents the average of the official soybean yield values published by the USDA.

R^2 measures how well the model explains the variance in the actual data, with higher values indicating a better fit. rRMSE is a normalized version of the root mean squared error, providing insight into the size of prediction errors relative to the data range, where smaller values suggest better model accuracy. MAPE calculates the average percentage error between predicted and actual values, with lower values indicating higher accuracy. rBias assesses the average tendency of the model to overestimate or underestimate values, helping identify systematic errors.

3. Results

3.1. Correlation of soybean yield with environmental and remote sensing variables

As in Fig. 4, from 2019 to 2022, the correlation between soybean yield at the county level in the U.S. and remote sensing variables (NDVI, EVI, NIRv, SIF, GPP) was found to be stronger than with environmental variables (mean temperature, maximum temperature, SM 0–10 cm, SM 10–40 cm). The correlation between yield and variables such as LAI, Temp_min, and precipitation was relatively weak, with Pearson correlation coefficients being close to zero for the entire growing season.

From June 25 to August 29, the correlation between variables and yield was significantly higher than during other periods. Specifically, the Pearson correlation coefficients for vegetation indices (NDVI, EVI, NIRv) ranged from 0.6 to 0.7, whereas in the early growth stages, the correlation between remote sensing variables and yield was nearly zero, with a slight negative correlation trend. Notably, compared to NDVI, NIRv exhibited a higher correlation with yield in the later growth stages, suggesting that NIRv can more effectively capture variations in photosynthesis, particularly during key stages such as flowering and grain filling.

In the later growth stages, the Pearson correlation coefficients for SIF and GPP with yield were approximately 0.4, while in the early stages, the correlation coefficients were nearly zero, with a slight negative correlation trend similar to that observed for the vegetation indices. For temperature variables, the correlation coefficients in the later stages ranged from -0.2 to -0.4 , with both mean temperature and maximum temperature displaying a strong negative correlation trend. In the early growth stages, their correlation coefficients were nearly zero.

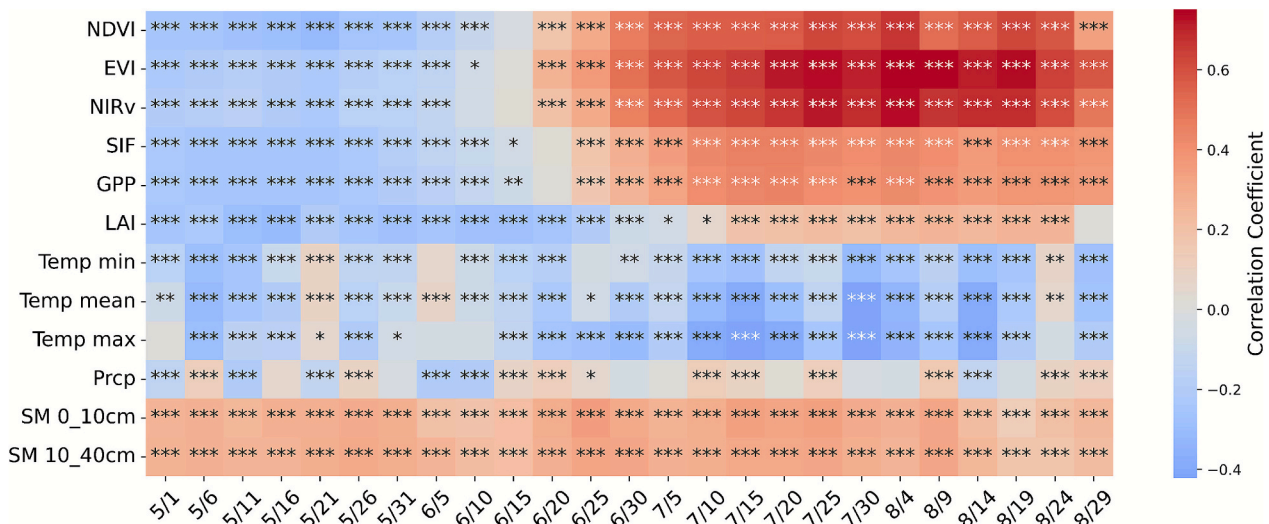


Fig. 4. Correlation of soybean yield with environmental and remote sensing variables in training dataset. The shade of color represents the magnitude of the correlation coefficient for different variables. * indicate a p -value between 0.01 and 0.05, suggesting the result is statistically significant. ** indicate a p -value between 0.001 and 0.01, suggesting the result is highly significant. *** indicate a p -value less than 0.001, suggesting the result is extremely significant.

Throughout the entire growing season, the correlation coefficients for soil moisture variables (SM 0–10 cm and SM 10–40 cm) remained relatively stable between 0.2 and 0.4, suggesting that soil moisture characteristics exhibit strong robustness in predicting yield.

Furthermore, from June 25 to August 29, the p -values for all selected features were below 0.001, confirming their statistical significance. Therefore, we selected the most relevant features for model training, validation, and testing: NDVI, EVI, NIRv, SIF, GPP, mean temperature, maximum temperature, SM 0–10 cm, and SM 10–40 cm.

3.2. Soybean yield prediction using environmental and remote sensing variables

We first evaluated the yield prediction performance of various models using different combinations of remote sensing and meteorological variables. The scatter plots of the prediction results for each model and feature combination are presented in Fig. 5, and the accuracy metrics in bar charts are shown in Fig. 6. Parameter optimization for each combination and model, through cross-validation, can be found in Supplementary Material Table B–D. The optimal parameters show that the RF model best parameters are relatively stable regardless of the input feature combinations. The XGBoost model exhibits slightly more variability, particularly in learning rate and tree depth, suggesting some adaptation to different input sets. In contrast, the AGB-LSTM model shows the most significant changes in optimal parameters across varying input combinations (like batch size, learning rate, and hidden layer sizes). This indicates that the more complex AGB-LSTM model dynamically adjusts its internal configuration to best utilize the specific information present in different sets of high-temporal-resolution input features, showcasing its adaptability. The results demonstrated significant variations in model performance. The linear model had the lowest prediction accuracy, while tree-based models showed similar results. The AGB-LSTM model achieved the highest accuracy ($R^2 = 0.65$, rRMSE = 15 %). Among feature combinations, VI + SIF + GPP delivered the best performance. Scatter plots indicated that predictions from all models closely followed the 1:1 line, without apparent non-linear or heteroscedastic patterns. Tree-based models exhibited slopes closer to 1 but with a discernible intercept, suggesting overestimation in both low- and high-yield areas. The AGB-LSTM model aligned closely with the 1:1 line in high-yield areas, suggesting reliable predictions in regions with larger sample sizes. Compared to the LR, RF, XGBoost, Transformer, AB-LSTM, and Bi-LSTM models, the AGB-LSTM model showed significant

improvement in performance.

When environmental variables, especially temperature, were included, linear models not only matched but sometimes even outperformed tree-based models. The AGB-LSTM model consistently demonstrated the highest prediction accuracy, especially with the VI + SIF + GPP + TEMP combination ($R^2 = 0.67$, rRMSE = 14 %). Scatter plots revealed that with the inclusion of temperature, the yield estimates from the AGB-LSTM model closely aligned with the 1:1 line, indicating minimal non-linearity or heteroscedasticity. This suggests that incorporating temperature significantly enhanced prediction accuracy across regions with varying sample sizes, surpassing both LR and tree-based models. Specifically, compared to LR, AGB-LSTM exhibited a year-on-year increase in R^2 by 30.4 %, a reduction in rRMSE by 15.6 %, a decrease in MAPE by approximately 5 %, and a lowering of rBias by about 8 %. In comparison to tree-based models, R^2 increased by 22.5 % year-on-year, rRMSE decreased by 14 % year-on-year, MAPE dropped by approximately 3 %, and rBias decreased by about 5 %. Compared to the transformer model, the AGB-LSTM model demonstrated an increase in R^2 by approximately 5 % to 10 % across all input feature combinations, accompanied by an average reduction of about 4 % in rBias. Relative to the AB-LSTM model, AGB-LSTM exhibited an increase in R^2 by approximately 5 % across all input feature combinations, with an average reduction of about 3 % in rBias. The rBias value consistently remained below 5 %, indicating that the predictions of our model did not exhibit any significant systematic bias. These improvements in accuracy are consistent with the findings observed when using only remote sensing variables.

To compare and validate the models, we employed three traditional machine learning algorithms (LR, RF, XGBoost, Transformer) for performance evaluation. As in Fig. 7, the absolute errors of the AGB-LSTM model were consistently significantly lower than those of the machine learning models (LR, RF, XGBoost, Transformer) for all feature combinations (a–h). Furthermore, the AGB-LSTM model outperformed the deep learning models (Bi-LSTM, AB-LSTM) in terms of absolute error in the majority of feature combinations (b–g). The boxplot illustrates that the median error of AGB-LSTM is lower with a smaller interquartile range, indicating that its predictions are not only more accurate but also more stable. In contrast, traditional models, particularly LR, exhibited higher errors and greater variability in their distributions. The significance test results demonstrate that the error differences between AGB-LSTM and other models are statistically significant for most feature combinations, emphasizing its superior accuracy and robustness. This is

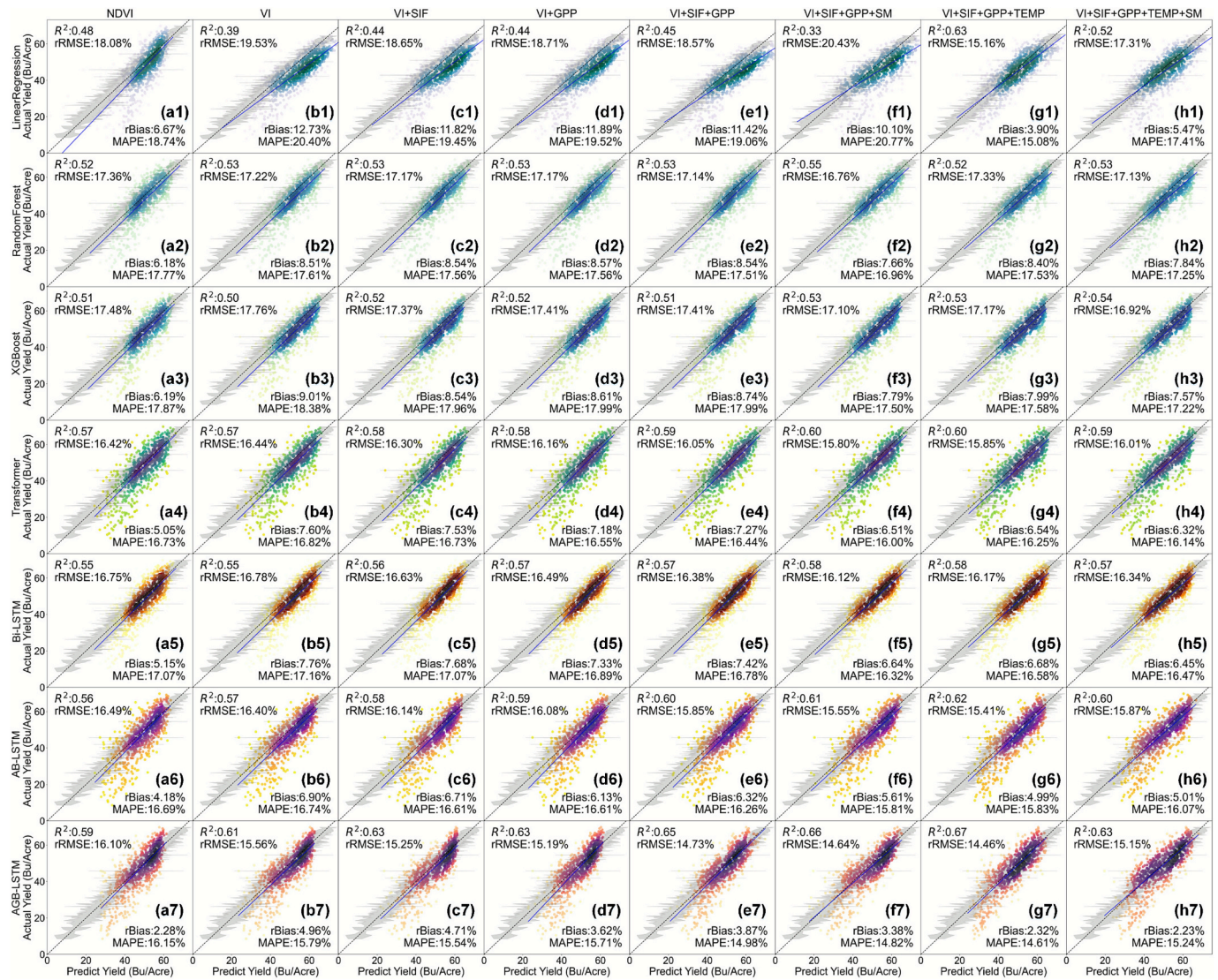


Fig. 5. Comparisons between the predicted soybean yield and the USDA official soybean yield data for 687 counties in the U.S. The gray threshold indicates CI. The blue line represents the fitted line of the model, and the color intensity of the scatter points reflects their density.

because linear models assume a linear relationship between input variables and yield. Their lowest prediction accuracy indicates a distinctly non-linear relationship between remote sensing data, environmental factors, and county-level soybean yield, which linear models struggle to capture along with the complex interactions and thresholds controlling crop growth. Tree-based models (RF, XGBoost) are more adept at capturing non-linear relationships by partitioning the feature space than linear models. Their superior performance over linear models confirms the presence of non-linearity. However, their results consistently show a significant bias in low-yield areas. This suggests that while they can capture non-linear patterns, their piecewise approximation nature or specific ensemble mechanisms may not perfectly model the yield distribution across the entire range, leading to systematic errors at extreme values. AGB-LSTM and other sequence models (Transformer, AB-LSTM, Bi-LSTM), particularly AGB-LSTM, consistently exhibit the highest accuracy. This suggests that the Graph Isomorphism Network and attention mechanism endow it with superior capabilities in weighting important features and capturing complex interrelationships between different nodes, surpassing even other sequence-aware models like LSTM, Bi-LSTM and Transformer. The performance improvement of AGB-LSTM is particularly significant when incorporating environmental variables such as temperature, highlighting its ability to effectively

integrate and utilize various data streams to enhance prediction accuracy. This suggests that AGB-LSTM is more suitable for prediction tasks that demand high precision and stability.

In this study, we focused on major soybean production areas in the U. S. Midwest (Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, and South Dakota) to map county-level yield predictions. Fig. 8 indicates that the impact of different models on prediction performance is notable. Using models such as RF (MAE = 6.23) and transformer (MAE = 5.74) resulted in lower prediction accuracy and larger errors, particularly in Missouri and Minnesota, where there was a noticeable overestimation. These regions correspond to counties with lower yields in the yield maps, which is consistent with the predictions shown in Fig. 5. However, the use of deep learning models effectively improved model performance. The best prediction results, obtained using the AGB-LSTM model (MAE = 5.54), demonstrated that county-level yield predictions were highly consistent with the official USDA values, with a reduction in error distribution and improvements in prediction accuracy for both low and high yield areas. This highlights that integrating multi-source remote sensing and environmental variables can effectively capture the spatiotemporal variability of soybean yield, providing stable predictions for high-yield regions while reducing errors in low-yield areas.

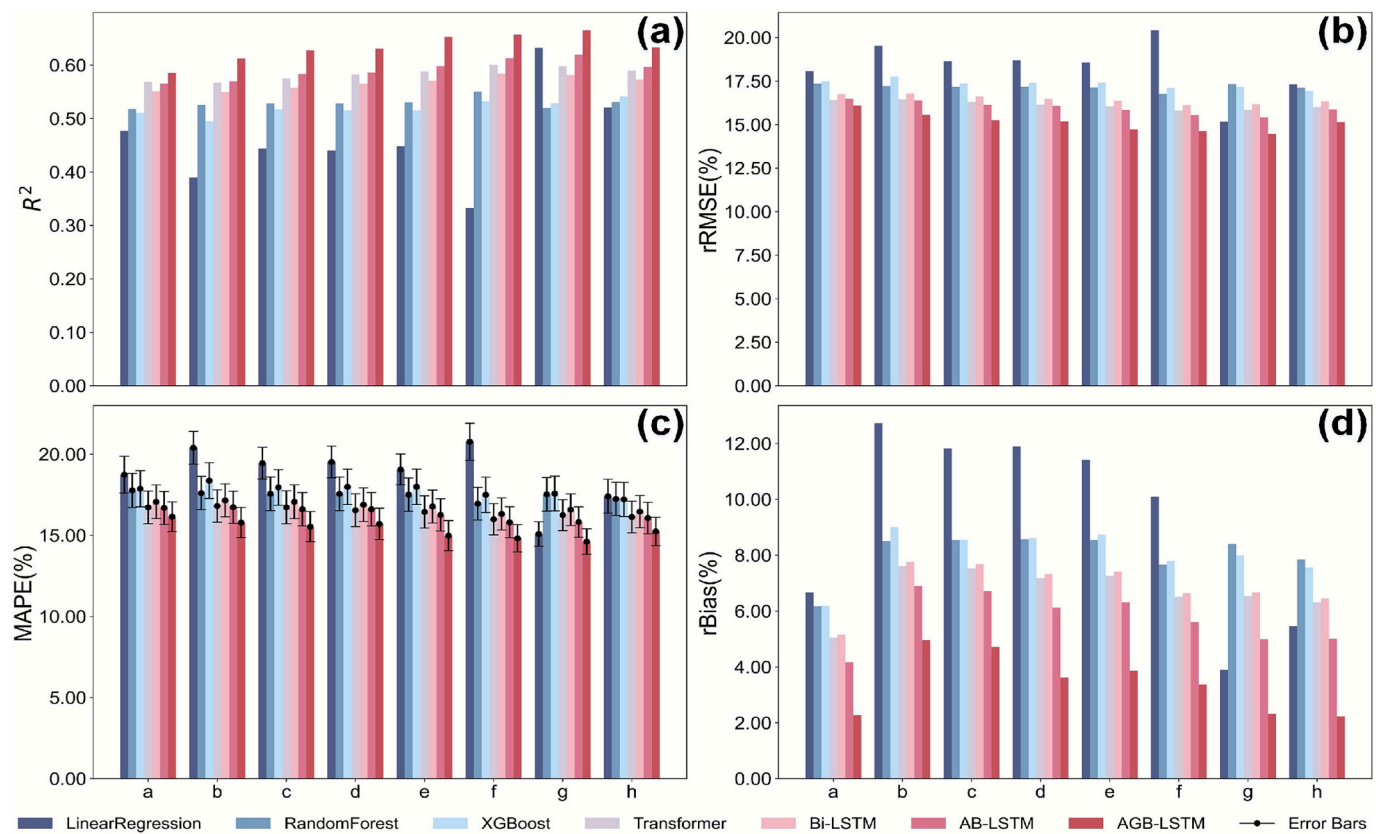


Fig. 6. Performance of soybean yield prediction with different models and feature combinations. The scale on the x-axis from a to h represents the feature combinations of NDVI, VI, VI + SIF, VI + GPP, VI + SIF + GPP, VI + SIF + GPP + SM, VI + SIF + GPP + TEMP, and VI + SIF + GPP + TEMP + SM.

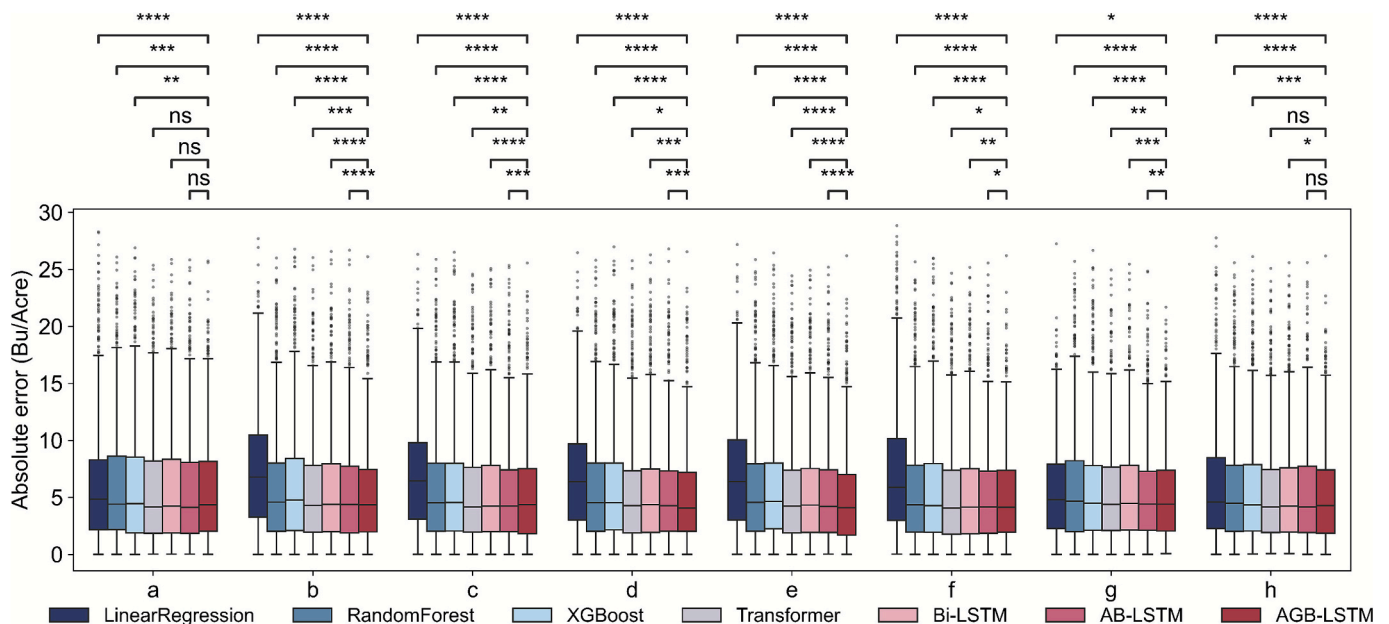


Fig. 7. Absolute prediction errors of different yield models under different feature combinations. The absolute prediction error is defined as the absolute difference between the predicted yield and the actual yield reported by USDA. Subfigures (a)-(h) represent the absolute errors of prediction results for the feature combinations of NDVI, VI, VI + SIF, VI + GPP, VI + SIF + GPP, VI + SIF + GPP + SM, VI + SIF + GPP + TEMP, and VI + SIF + GPP + TEMP + SM, respectively. Here, ns: $p > 0.05$, *: $0.01 < p < 0.05$, **: $0.001 < p < 0.01$, ***: $0.0001 < p < 0.001$, ****: $p < 0.0001$.

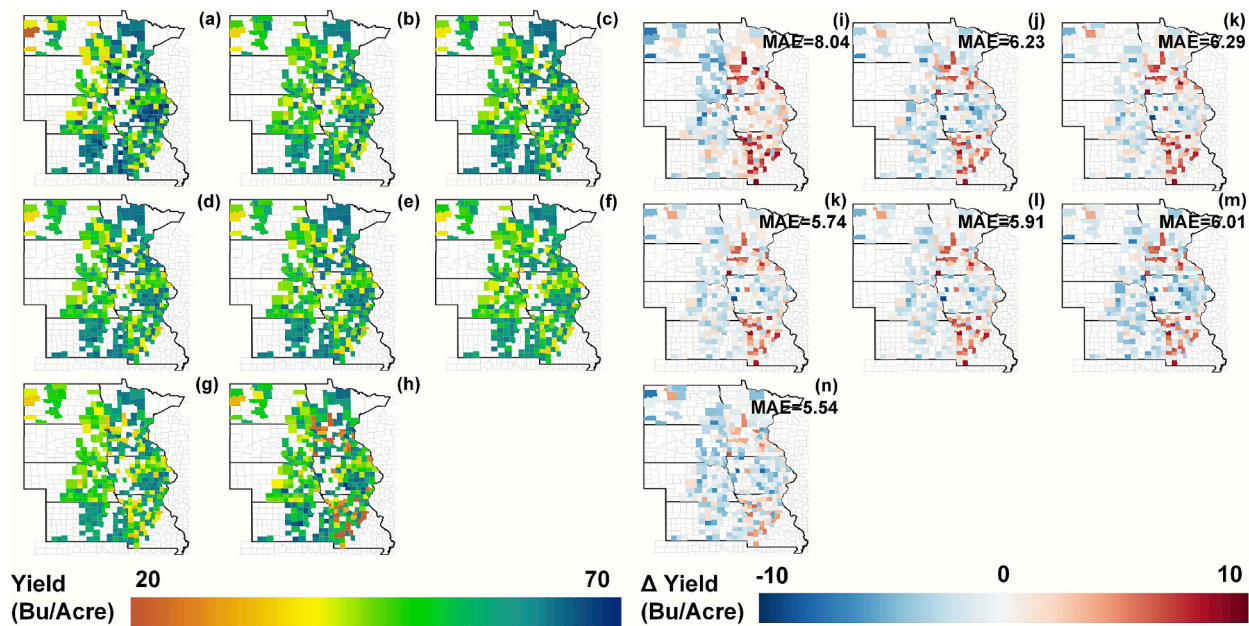


Fig. 8. Distribution and error map of soybean yield predictions in major soybean producing areas of the USA, 2023. Subfigures (a)-(h) and (i)-(n) represent the prediction results of different models: LR, RF, XGBoost, Transformer, Bi-LSTM, AB-LSTM and AGB-SLTM. Subfigures (a)-(g) show the county-level soybean yield estimates for the major soybean-producing states in the U.S. (Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, and South Dakota) in 2023. Subfigure (h) represents the true values of soybean production at the county level in the U.S. in 2023 released by USDA. Subfigures (i)-(n) illustrate the differences (Δ) between the prediction results for each model and the official USDA data.

3.3. In-season prediction of soybean yield and evaluation of time step impact

As in Fig. 9, the results of the in-season predictions suggest that our model can achieve considerable predictive accuracy as early as the flowering stage of soybean crops (late June). Both the rRMSE and MAPE are no greater than 20 %, with the rBias close to 1 %, indicating that our model can accurately predict over 80 % of soybean yield during the flowering period. By the pod-setting stage (early to mid-July), the predictive performance of the model improves further, with an R^2 of approximately 0.5, and both the rRMSE and MAPE reaching about 17 %.

During the pod-filling stage (mid-July to mid-August), the prediction performance of the model improves markedly and gradually stabilizes, with the highest R^2 exceeding 0.6, and both rRMSE and MAPE reaching around 15 %. At the soybean maturity stage (mid-August to early September), the accuracy of our model peaks ($R^2 = 0.67$, rRMSE = 14 %). Temporally, the accuracy metrics exhibit a clear monotonic trend: R^2 increases monotonically over time, while both rRMSE and MAPE decrease monotonically and stabilize during the pod-filling stage (early August). This indicates that we can predict soybean yield with high accuracy up to two months prior to the harvest period (early September to early October). Although the rBias fluctuates to some extent, its base

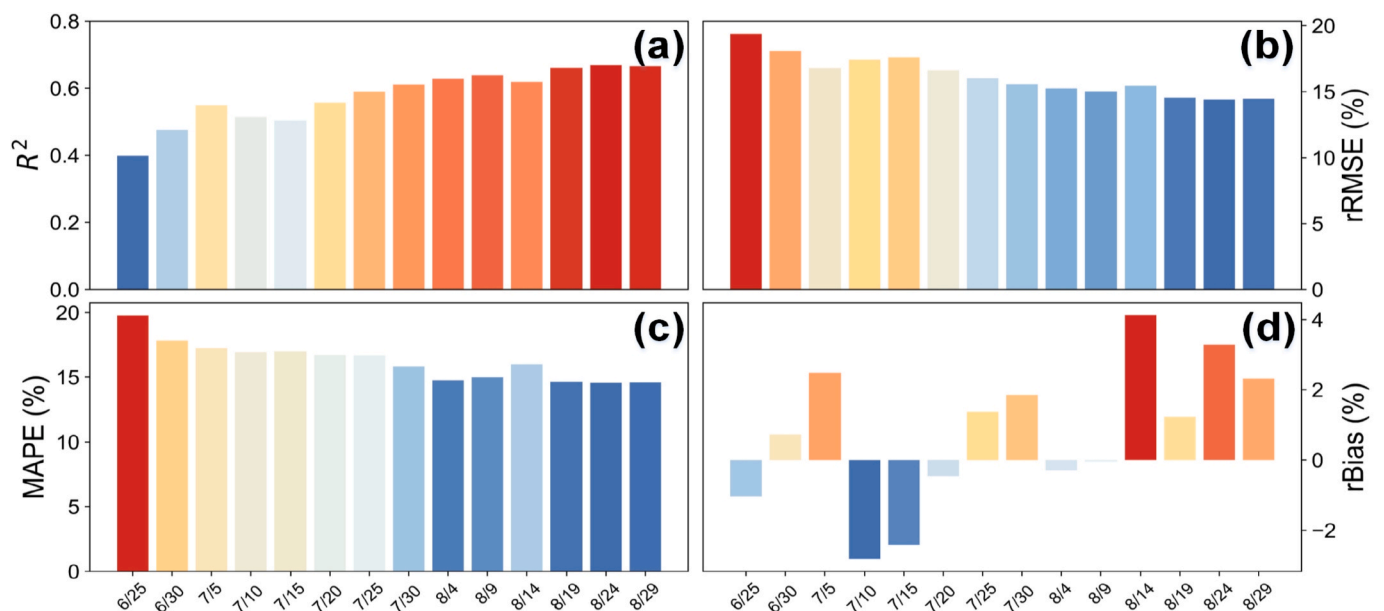


Fig. 9. The bar charts of soybean yield prediction accuracy metrics over time for early prediction: Subfigures (a) to (d) represent the accuracy results for R^2 , rRMSE, MAPE, and rBias, respectively. The shading of the bars indicates the magnitude of the values on the y-axis.

value remains small, and throughout the study period, the absolute value of rBias does not exceed 5 %, further demonstrating the robustness of our model.

Fig. 10 illustrates the forecast results of sequential data over different time step lengths, revealing notable variations in the prediction accuracy of soybean yield per unit area. As the time step decreases from 30 days to 5 days, the prediction accuracy of the model demonstrates a monotonic increase, albeit at a decreasing rate. Specifically, R^2 improves from 0.55 at a 30-day time step to 0.67 at a 5-day time step, indicating that shorter time steps more effectively capture variations in actual yields. Correspondingly, rRMSE decreases from 17 % to 14 %, signifying a substantial reduction in prediction errors. Additionally, MAPE also progressively decreases with shorter time steps, reducing from 17.40 % at 30 days to 14.61 % at 5 days, and rBias drops from 6.59 % to 2.32 %, suggesting that models with shorter time steps more closely approximate actual values. The density distribution in the figure reveals that predictions at a 5-day time step align more closely with actual values, with most points concentrated near the ideal prediction line, whereas the 30-day time step shows a more dispersed point distribution, deviating more from the ideal line, especially in low-yielding areas (less than 40 bu/acre). Thus, predictions with shorter time steps exhibit superior accuracy and reliability, particularly suitable for scenarios requiring high precision, while a 30-day time step might be more appropriate for long-term trend predictions or situations with lower accuracy requirements. Overall, shorter time steps more effectively capture detailed variations, thereby enhancing model performance. However, the improvement in accuracy metrics from a 10-day to a 5-day time step does not exceed 5 %, indicating that the model performance gradually reaches saturation with finer time steps.

3.4. The impact of extreme events on yield prediction

Fig. 11 presents the results of extreme climate event detection. The analysis reveals clear temporal patterns in the occurrence of extreme climate events, including precipitation, extreme high temperatures, and extreme low temperatures. Regarding precipitation, extreme precipitation events exhibit distinct peaks on May 6, June 15–July 15, and around August 29, notably exceeding the median value and the 95th percentile upper limit of the 2019–2022 baseline period. This elevated precipitation poses a notable risk of flooding. Notably, between May 21 and June 5, the median value of extreme precipitation remained nearly constant and close to 0 mm, possibly indicating a severe drought event. In terms of temperature, extreme high-temperature events persisted above average from mid-July to mid-August, culminating in a peak on August 19, with temperatures exceeding 40 °C, notably higher than the baseline range. This observation suggests an increase in the intensity and frequency of high-temperature events, potentially leading to heatwaves. Conversely, extreme low-temperature events were prominent in mid-June and late August, particularly around June 10. It is noteworthy that the frequency of warm extreme low-temperature events and cold extreme low-temperature events appears to be comparable.

Furthermore, the overall median line for extreme high and low temperatures lies above the baseline, indicating a general warming trend.

We considered as many environmental variables as possible, including temperature and soil moisture, to enhance the robustness of the model even under extreme climate conditions. As in Fig. 12, the model demonstrates good prediction accuracy and stability in samples without extreme climate events, with a high correlation between actual and predicted yields and a small prediction bias ($R^2 = 0.64$, rRMSE = 15 %). However, in samples affected by extreme climate events, the performance of the model degrades, with reduced correlation ($R^2 = 0.50$, rRMSE = 21 %) and increased dispersion of data points. Some samples deviate from the 1:1 line, indicating the complex and nonlinear impact of extreme climate events on yield, which in turn increases the uncertainty in predictions. Notably, the model proposed in this study achieved the highest prediction accuracy among the models we tested with or without extreme climate impacts, which demonstrates the robustness of our model. Despite this, most samples from high-yield areas still fall within the acceptable error threshold, with only a few low-yield samples being overestimated. The MAPE is approximately 23 %, suggesting that the absolute prediction accuracy for most samples exceeds 75 %, while the rBias remains under 5 %. This indicates that our model maintains a certain level of robustness, even under the influence of extreme climate events.

4. Discussion

4.1. Role of remote sensing measures in soybean yield prediction

Remote sensing data, especially spectral reflectance and vegetation indices, effectively reflect crop growth conditions and provide accurate estimates of biophysical characteristics across growth stages (Zeng et al., 2022a; Zhang et al., 2025a, b; Liu et al., 2025). The results of this study indicate that key reproductive stages of soybean, such as the pod expansion and maturity stages, are crucial for determining seed quantity and quality (Fig. 4). This finding is consistent with that of Torrion et al. (2011). The feature importance analysis (Fig. 4) reveals that during this period, the correlation between vegetation indices and soybean yield significantly increased, particularly for EVI and NIRv. NIRv demonstrated considerable potential in yield prediction, aligning with the findings of Peng et al. (2020b). Compared to traditional vegetation indices, NIRv approximates the ratio of near-infrared (NIR) light reflected by the plant canopy, minimizing the influence of soil background and sensor geometry (Dechant et al., 2022). Numerous studies have demonstrated that NIRv serves as a good proxy for soybean GPP (Dechant et al., 2022; Wu et al., 2020b). Furthermore, the results in Fig. 5 show that incorporating SIF or GPP as additional feature variables still yields a positive impact on model prediction performance, even when NIRv is already included. This suggests that there is not a completely substitutive relationship between NIRv, SIF, and GPP. For example, advanced research by He et al. (2020) has demonstrated that C3 crops, such as soybeans, exhibit GPP saturation under high light

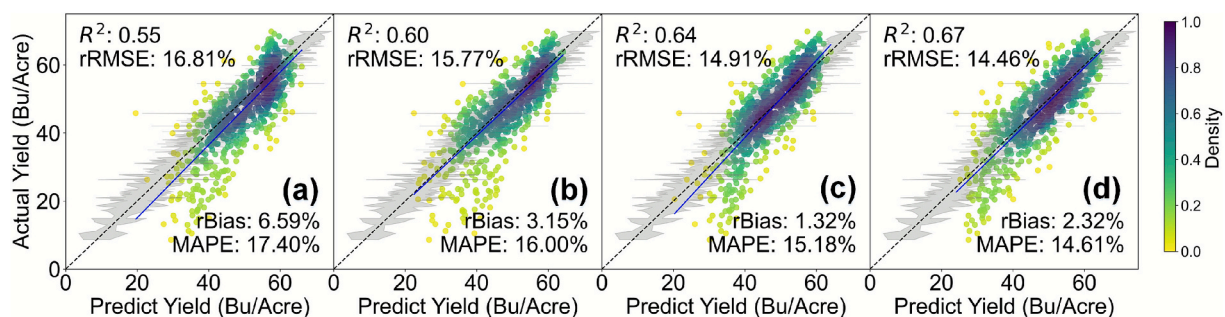


Fig. 10. Comparisons between the predicted soybean yield and the USDA official soybean yield data, based on sequence data with different time step lengths. Subfigures (a) to (d) represent the prediction results for time steps of 30 days, 15 days, 10 days, and 5 days, respectively.

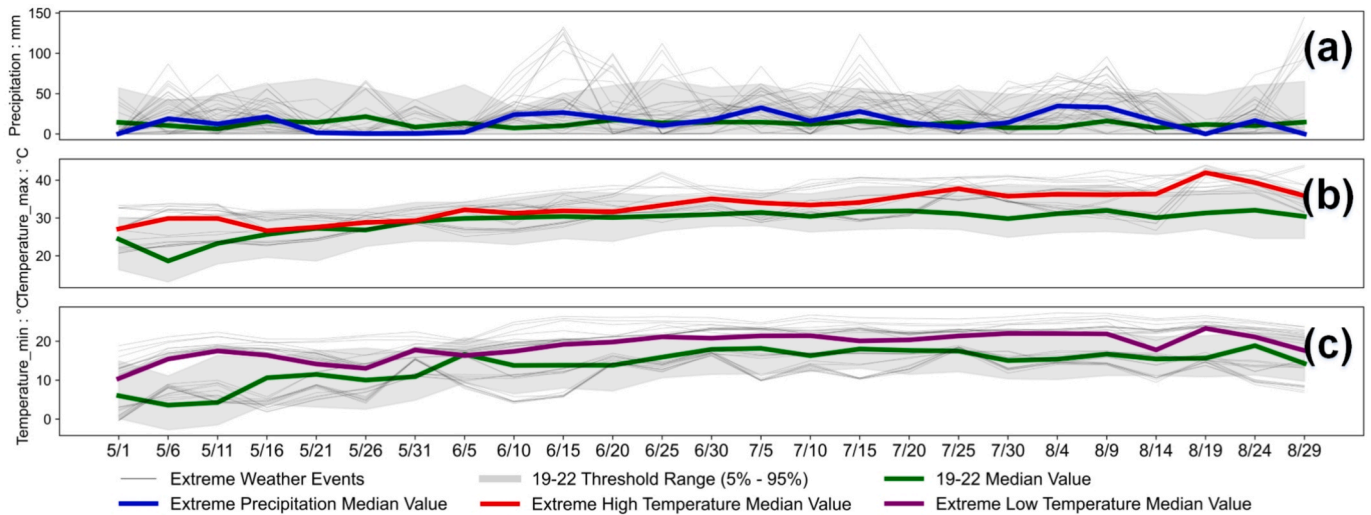


Fig. 11. Extreme climate events examined using the RCLIMDEX model. Panel (a) shows extreme precipitation events, panel (b) shows extreme high-temperature events, and panel (c) shows extreme low-temperature events.

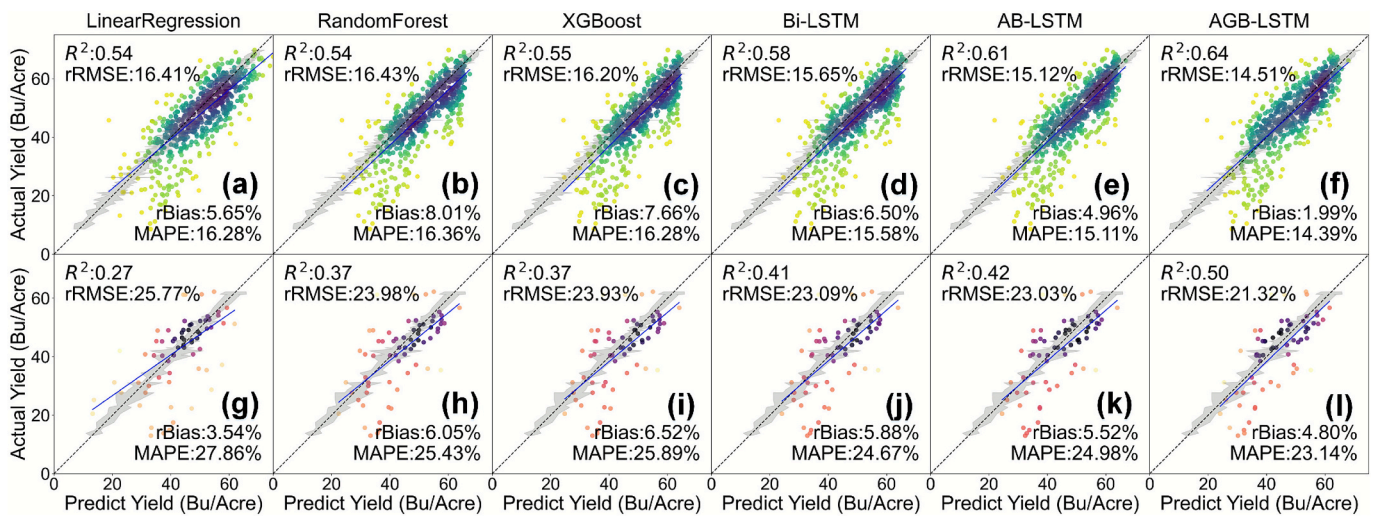


Fig. 12. Comparisons between the predicted soybean yield and the USDA official soybean yield data, under conditions with and without extreme climate events. Subfigures (a) to (f) indicate the absence of extreme climate events, while subfigures (g) to (l) indicate the presence of extreme climate events.

conditions, while SIF values continue to increase. Future research could explore the interactions among NIRv, SIF signals, and GPP to develop more suitable indicators for soybean yield estimation.

Furthermore, future work may benefit from the use of higher-quality SIF data. On one hand, the SIF signal at the top of the soybean canopy can be converted into total SIF emission, which could better relate to the photosynthetic activity of the soybean canopy (Zeng et al., 2019). Zeng et al. (2019) proposed a method using the NIRv-fPAR ratio to approximate the near-infrared SIF escape ratio, enabling the large-scale conversion of satellite-observed SIF to total SIF. On the other hand, using SIF products from airborne fluorescence sensors or employing advanced super-resolution networks for downscaling can improve the accuracy of yield predictions (Liu et al., 2019).

In addition, although SIF is related to plant photosynthesis, it is also influenced by factors such as the atmosphere, leaf structure, and lighting conditions. Therefore, fluorescence yield (Φ_F) theoretically provides a more direct reflection of the physiological state of the plant. Using Φ_F can also contribute to more accurate and efficient crop yield estimation. Zeng et al. (2022b) proposed a method for normalizing SIF to the canopy structure using NIRvR, and extracting Φ_F variations at the canopy scale. This approach effectively captures the diurnal and seasonal variations of

Φ_F without being influenced by canopy structure and sun-sensor geometry.

4.2. Importance of environmental factors in soybean yield prediction

Environmental and remote sensing data provide independent and complementary information for soybean yield prediction (Fig. 5). On the one hand, although the correlation between environmental features and yield may not be as direct as that with remote sensing data, they exert a sustained and notable influence on soybean yield throughout the growing season and can substantially improve the accuracy of predictions solely based on remote sensing data. This finding is consistent with the results of Mourtzinis et al. (2015). On the other hand, the results in Fig. 11 show that during the critical growth stages of soybean, both low and high-temperature curves are notably higher than the historical baseline curves, and the frequency of extreme high-temperature and drought events is markedly higher than that of extreme low-temperature events. This confirms that extreme high temperatures and drought-heat disasters are the primary factors influencing crop yield, which aligns with the findings of Leng and Hall (2019).

The impact of environmental factors and extreme climate events

should be further explored in future studies. On the one hand, we computed the average of environmental data at the county level. However, this approach may not accurately reflect the actual environmental conditions of the target soybean fields. Future research could benefit from utilizing more precise environmental data obtained through observation methods such as the Internet of Things (IoT) or tower-based measurements, which would likely enhance the accuracy of crop yield estimations. On the other hand, the presence of extreme climate events introduced a relatively high degree of uncertainty into our model. Although environmental features were incorporated during the training process, samples influenced by extreme climate events still represented only a small proportion of the overall dataset, posing a notable challenge for data-driven deep learning models (Johnson and Khoshgoftaar, 2019). One potential solution involves implementing pre-discriminatory constraints and separately predicting yields for samples affected by extreme climate events and those unaffected. Another promising approach is to integrate deep learning models with process-based crop growth models (Shahhosseini et al., 2021). By simulating various environmental conditions, including extreme events, to model the yield response of the crop, the resulting data can be used as input to improve the generalization performance of the deep learning model.

4.3. Limitations

Despite its high accuracy and stability, the proposed model exhibits a tendency to overestimate yields in low-yield areas. This phenomenon likely arises from the inherent requirement of deep learning models for large and diverse datasets. Low-yield areas, by their nature, tend to have fewer samples in the official USDA county-level statistics, resulting in an imbalanced dataset that hinders the ability of the model to effectively learn their characteristics. Direct data augmentation on such aggregated data without field-level granularity risks introducing synthetic noise that could compromise validity. This finding is consistent with the conclusions of Johnson and Khoshgoftaar (2019).

Additionally, noise arising from deep neural network architectures, network parameters, and input data can introduce certain biases into the prediction results (Tian et al., 2024). Additionally, although the USDA-CDL provides relatively accurate crop classification maps, obtaining up-to-date crop classification products for the current year can be challenging in practical yield estimation studies.

Moreover, considering the prevalence of multi-cropping systems with various crops throughout the year in fields across the Southern U. S., this implies that the critical growth stages of soybeans may differ compared to fields with single cropping per year. The transferability of our model has not yet been effectively validated against this temporal heterogeneity. While the model demonstrates good performance at the county level, its performance at the field or pixel level has not yet been validated.

4.4. Prospects

To address the problem of overestimation of production forecasts, several strategies can be implemented: (1) Employing data augmentation techniques to increase the sample size and diversity within low-yield areas, and existing samples were augmented by conducting additional drone field experiments. (2) Implementing a weighted loss function during model training to assign higher penalties to samples from low-yield areas and mitigate the impact of regional imbalance; (3) Integrating external variables, such as geographic factors and soil nutrient levels, to enhance the ability of the model to capture the unique features of low-yield areas and improve prediction accuracy.

To solve the problems of network architecture and data noise, several strategies can be implemented: (1) Multi-source data fusion and denoising techniques can be employed to enhance the precision of data collection and improve data quality. For example, Maimaitijiang et al. (2020) estimated soybean yield within a deep neural network (DNN)

framework by extracting multi-modal data from drones, integrating canopy spectra, crop phenotypes, thermal infrared, and texture features. Future research could further incorporate hyperspectral drone imagery and synthetic aperture radar (SAR) data. The use of satellite data across different spectral ranges can notably enhance the monitoring of large-scale crop growth and yield, surpassing the capabilities of a single sensor (Guan et al., 2017). Specifically, hyperspectral data provides high spectral resolution, offering valuable insights into the biophysical state and growth status of the crop, while SAR data can effectively complement dynamic changes in crop structure and soil moisture, thereby enabling more accurate yield estimation (Li et al., 2020; Sah et al., 2024). (2) Conducting real-time or near-real-time crop classification during the growing season is crucial. Future research should prioritize improving the accuracy of crop classification to mitigate data uncertainty and consequently achieve more accurate yield estimates.

Considering that crop rotation is commonly practiced in the U.S., such as the rotation between soybean and corn, it is necessary to perform multi-year yield estimation over a longer time frame to assess the accuracy of yield predictions under crop rotation patterns (Lawes et al., 2022). Future research will focus on quantifying the impact of crop rotation systems on soybean phenology.

Moreover, integrating advanced deep learning networks would allow future research to evaluate the accuracy of the model at the field scale. For instance, Vafaeinejad et al. (2025) proposed a fully automated method utilizing the “Segment Anything Model” (SAM) for the digital extraction of agricultural cadastral maps. By enabling the automatic generation of these maps, this research signifies a major advancement for land management systems, addressing the technical and operational challenges associated with map updating and digitization. Ma et al. (2024) developed a Quantile Loss Domain Adversarial Neural Network (QDANN) and generated 30-meter yield maps for the U.S. Corn Belt and wheat belt from 2008 to 2022, providing a reliable method for field-scale yield mapping. Future research could focus on improving crop classification accuracy to minimize noise and uncertainty in the input data, thereby achieving better yield estimation.

5. Conclusions

In this study, we proposed an Attention and Graph Isomorphism Network-enhanced Bi-directional Long Short-Term Memory network (AGB-LSTM), and integrate vegetation indices, SIF, GPP, LAI, and environmental factors to predict county-level soybean yields in the U.S. The proposed model effectively extracts information related to crop yield from high-temporal-resolution time series data (5 days), achieving an independent validation accuracy ($R^2 = 0.67$, $rRMSE = 14.46\%$) that significantly outperformed traditional machine learning methods. Experimental results demonstrate that our model exhibited significantly improved predictive performance and achieved gradual stabilization during the pod-filling period (mid-July to mid-August). Furthermore, higher temporal resolution data enhanced the prediction performance. Despite the influence of extreme climatic events, our model demonstrated robust predictive performance ($R^2 = 0.50$, $rRMSE = 21.32\%$). Finally, yield mapping for major soybean-producing regions in North America in 2023 reveals spatial patterns that closely match USDA yield reports. Our findings suggest that the AGB-LSTM model is a promising and effective method for yield estimation and demonstrates notable potential for global crop yield forecasting.

CRedit authorship contribution statement

Jia Yin: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Rundong Zhang:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Yelu Zeng:** Writing – review & editing, Methodology. **Peng Zhu:** Writing – review & editing, Methodology. **Leikun Yin:** Writing – review & editing, Methodology. **Yuchi Ma:** Writing – review &

editing, Methodology. **Wei Su:** Investigation. **Jianxi Huang:** Investigation. **Xuecao Li:** Investigation. **Dalei Hao:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2025.111283>.

Data availability

The AGB-LSTM framework is publicly available at <https://github.com/MITAKATO/AGB-LSTM.git>.

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